Negating Matchers

In *Putting the Pieces Together*, on page 172, we saw RSpec's not_to and to_not methods, which specify that a given condition should *not* hold:

12-creating-custom-matchers/05/custom_matchers.rb
```
expect(correct_grammar).to_not split_infinitives
```

If you find yourself doing this over and over again, you can define a *negated matcher* that you'd use like so:

12-creating-custom-matchers/05/custom_matchers.rb
```
expect(correct_grammar).to avoid_splitting_infinitives
```

It's easy to create your own negated matcher. All you have to do is call define_negated_matcher:

12-creating-custom-matchers/05/custom_matchers.rb
```
RSpec::Matchers.define_negated_matcher :avoid_splitting_infinitives,
                                       :split_infinitives
```

As with alias_matcher, you pass the name of the new matcher, followed by the old one. The avoid_splitting_infinitives matcher will now behave as the negation of split_infinitives.

Negative matchers come in handy for more complex cases, such as when you're combining matchers. For example, the following expectation is ambiguous, and RSpec warns us of this problem:

```
>> expect(adverb).not_to start_with('a').and end_with('z')
NotImplementedError: `expect(...).not_to matcher.and matcher` is not ↩
supported, since it creates a bit of an ambiguity. Instead, define negated ↩
versions of whatever matchers you wish to negate with ↩
`RSpec::Matchers.define_negated_matcher` and use `expect(...).to ↩
matcher.and matcher`.
    « backtrace truncated »
```

The ambiguity is subtle: should the adverb "absolutely" match (because it satisfies the "does not end with *z*" condition)? Or should it fail to match (because it does not satisfy *both* conditions)?

RSpec's error message points out the ambiguity and suggests negated matchers as an alternative. Here's what those negated matchers might look like:

12-creating-custom-matchers/05/custom_matchers.rb
```
RSpec::Matchers.define_negated_matcher :start_with_something_besides,
                                       :start_with

RSpec::Matchers.define_negated_matcher :end_with_something_besides,
                                       :end_with
```